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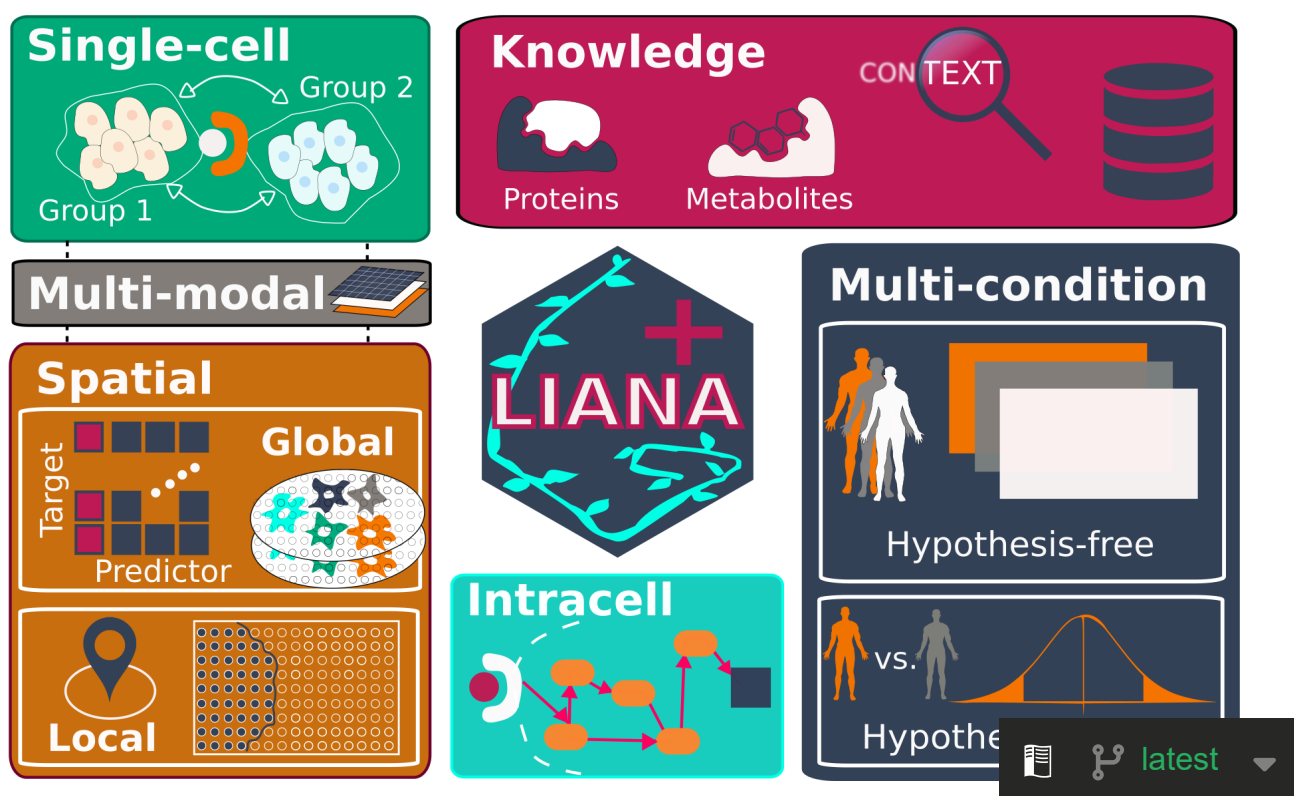
LIANA+: an all-in-one cell-cell communication framework

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LIANA+ is a scalable framework that adapts and extends existing methods and knowledge to study cell-cell communication in single-cell, spatially-resolved, and multi-modal omics data. It is part of the [scverse ecosystem](#), and relies on [AnnData](#) & [MuData](#) objects as input.



Contributions

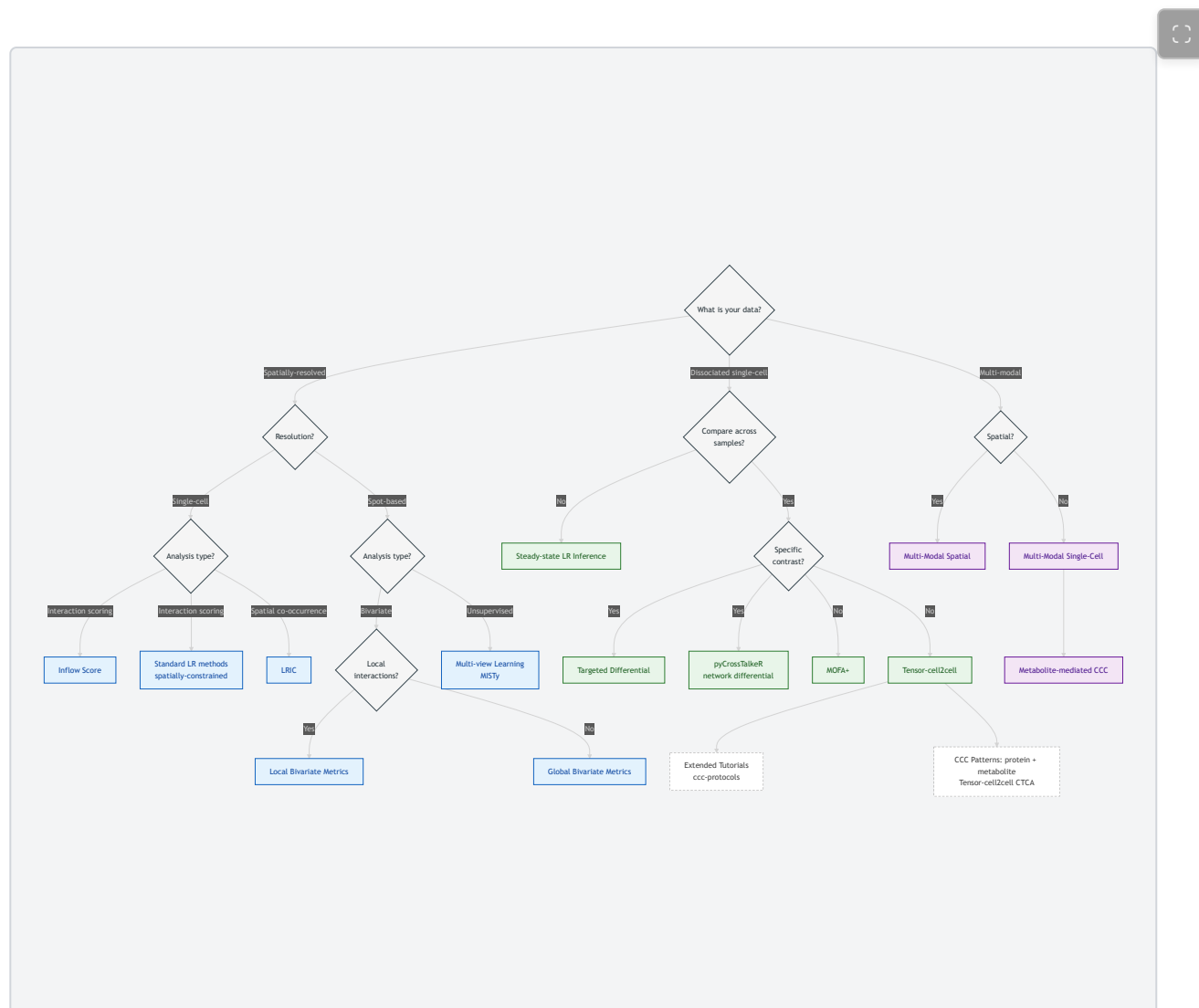
We welcome suggestions, ideas, and contributions! Please do not hesitate to contact us, open issues, and check the [contributions guide](#).

Vignettes

A set of extensive vignettes can be found in the [LIANA+ documentation](#).

Decision Tree

Use the tree below to find a starting point for your analysis. Broad, data-driven choices sit at the top and trickle down to specific methods (click a node to open its tutorial).



This tree is a guide rather than an exhaustive map: the methods are more adapted or combined across data types and questions, and all of them typically build on

curated prior knowledge (see the [prior knowledge](#) tutorial for working with ligand–receptor and other resources).

API

For further information please check LIANA's [API documentation](#).

Cite LIANA+:

```
@article {Dimitrov2024,
  author = {Dimitrov, Daniel and Sch{"a"}fer, Philipp Sven Lars and Farr, Elias and ...},
  title = {LIANA+ provides an all-in-one framework for cell--cell communication inference},
  journal = {Nature Cell Biology},
  year = {2024},
  volume = {26},
  number = {9},
  pages = {1613--1622},
  DOI = {10.1038/s41556-024-01469-w},
  URL = {https://doi.org/10.1038/s41556-024-01469-w}
}
```

```
@article {Dimitrov2022,
  author = {Dimitrov, Daniel and T{"u"}rei, D{"e"}nes and Garrido-Rodriguez, Mart{\'i}n},
  title = {Comparison of methods and resources for cell-cell communication inference},
  journal = {Nature Communications},
  year = {2022},
  volume = {13},
  number = {1},
  pages = {3224},
  DOI = {10.1038/s41467-022-30755-0},
  URL = {https://doi.org/10.1038/s41467-022-30755-0}
}
```

Please also consider citing any of the methods and/or resources that were particularly relevant for your research!



latest

